

Week 1 Discussion

Russell's paradox (1901)

In Week 1 we used sets freely: sample spaces, events, unions, complements. But **what is a set**, and can we form a set out of any description we like? At the turn of the 20th century, the answer was thought to be yes. Bertrand Russell showed it cannot be.

The paradox

“Naive” set theory (Cantor, Frege) allowed a set to be formed from **any** property φ :

$$\{x : \varphi(x)\}.$$

Russell took the property “ x is not a member of itself” and formed

$$R = \{x : x \notin x\}.$$

A popular version of the paradox: a barber shaves exactly those people who do not shave themselves. Who shaves the barber?

Now ask: **is** $R \in R$?

- If $R \in R$, then R does not satisfy the defining property, so $R \notin R$.
- If $R \notin R$, then R satisfies the defining property, so $R \in R$.

Both answers lead to their opposite, so naive set theory is **inconsistent**. ### How modern set theory avoids it

Modern mathematics, including probability, rests on the axioms of **ZFC** (Zermelo–Fraenkel set theory with the Axiom of Choice). In ZFC, “set” is not defined; it is a primitive notion governed by axioms. Two of them rule out Russell’s construction.

1. **Separation (Specification)**. You cannot form $\{x : \varphi(x)\}$ out of nothing. You can only carve a subset out of a set A you already have:

$$\{x \in A : \varphi(x)\}.$$

Russell’s set becomes $R_A = \{x \in A : x \notin x\}$. This is a perfectly good set. Asking whether $R_A \in R_A$ now leads only to the conclusion that $R_A \notin A$, not to a contradiction. One consequence: **there is no “set of all sets.”**

2. **Foundation (Regularity)**. Every nonempty set A has an element disjoint from A . It follows that **no set is a member of itself**: $x \in x$ is impossible (apply the axiom to $\{x\}$).

Bottom line. Because we always fix S first and build events, complements, and sigma algebras from it, every set we use in this course is legitimate, and Russell’s paradox cannot arise. The set-theoretic subtlety that *does* matter in probability is a different one, **non-measurable sets**. Using the Axiom of Choice, one can construct subsets of $[0, 1]$ that cannot be assigned any sensible length (Vitali, 1905). That is why we restrict P to a sigma algebra instead of all subsets of S .

Discussion questions

1. Let $S = \{1, 2, 3\}$ and let $A = \{x \in S : x \notin x\}$. What is A ? Why is there no paradox here?
 2. Why must a complement always be taken relative to a sample space S ? What would go wrong if A^c meant “everything not in A ”?
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Problem 1 (From words to sets)

Let A , B , and C be events in a sample space S .

(a) Write each event below using only $\cup$, $\cap$, and complements of A , B , C :

1. at least one of A , B , C occurs;
2. none of them occurs;
3. exactly one of them occurs;
4. at most two of them occur.

(b) Use DeMorgan’s laws to show that your answers to (1) and (2) are complements of each other, and rewrite (4) as the complement of a simpler event.

(c) Check your answers on a die: $S = \{1, \dots, 6\}$, $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$, $C = \{2, 3, 4\}$. List the outcomes in each of the four events.

Hint for (3): “exactly one” is a union of three pieces, one for each event that occurs alone. Are the three pieces disjoint?

Problem 2 (Proof by double containment)

(a) Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$. Then give a one-line proof using $A \setminus B = A \cap B^c$ and DeMorgan’s laws.

(b) True or false: $(A \cup B) \setminus B = A$. Prove it, or give a counterexample and state the correct identity.

Problem 3 (Infinite unions and intersections)

(a) Compute each set, and justify your answer with the “for some n ” / “for all n ” definitions:

$$\bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1 - \frac{1}{n} \right], \quad \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n} \right), \quad \bigcap_{n=1}^{\infty} \left[0, 1 + \frac{1}{n} \right).$$

(b) Let $A_1, A_2, A_3, \dots$ be any events. Define

$$B_1 = A_1, \quad B_n = A_n \cap A_1^c \cap \dots \cap A_{n-1}^c, \quad n \geq 2.$$

In words, B_n is “ A_n occurs, but none of the earlier A_i do.”

1. Show that the B_n are pairwise disjoint.
2. Show that $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$.
3. Find the B_n for $A_n = [0, n]$ with $n = 1, 2, \dots$. Do they form a partition of $[0, \infty)$?

(c) (*Challenge*) Interpret the event $\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n$ in words. (Consider an outcome x : for which k does x belong to $\bigcup_{n \geq k} A_n$?)

Why this matters: part (b) is the “disjointification” step we will use in Thursday’s lecture to prove Boole’s inequality.